

PrivacyAkinator:

Articulating Key Privacy Design Decisions by
Answering LLM-Generated Multiple-choice Questions

I want to design an attendee attention tracking feature for a video conferencing app.



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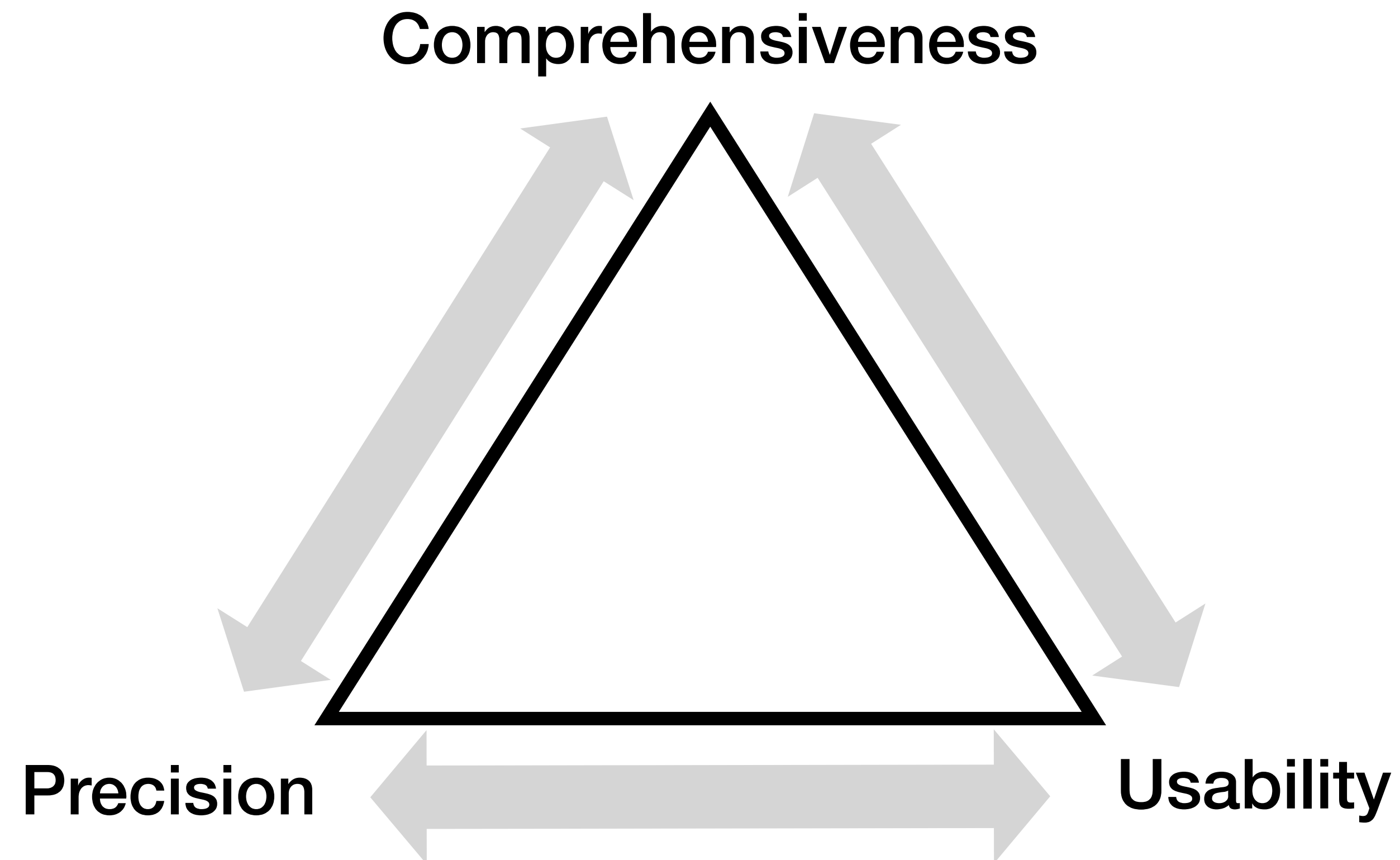
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Problem

Data Practice Reporting Trilemma



Data Collection

i **Contact Info**

☒ **Name**
Including first or last name

☒ **Email Address**
Including but not limited to a hashed email address

☐ **Phone Number**
Including but not limited to a hashed phone number

☐ **Physical Address**
Such as a home address, physical address, or mailing address

☐ **Other User Contact Info**
Any other information that can be used to contact the user outside the app

♥ **Health & Fitness**

☒ **Health**
Health and medical data, including but not limited to from the Clinical Health Records API, HealthKit API, MovementDisorderAPIs, health-related human subject research, or any other user provided health or medical data

☒ **Fitness**
Fitness and exercise data, including but not limited to the Motion and Fitness API

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Cancel

Save

Apple's app privacy questions

PrivacyAkinator:

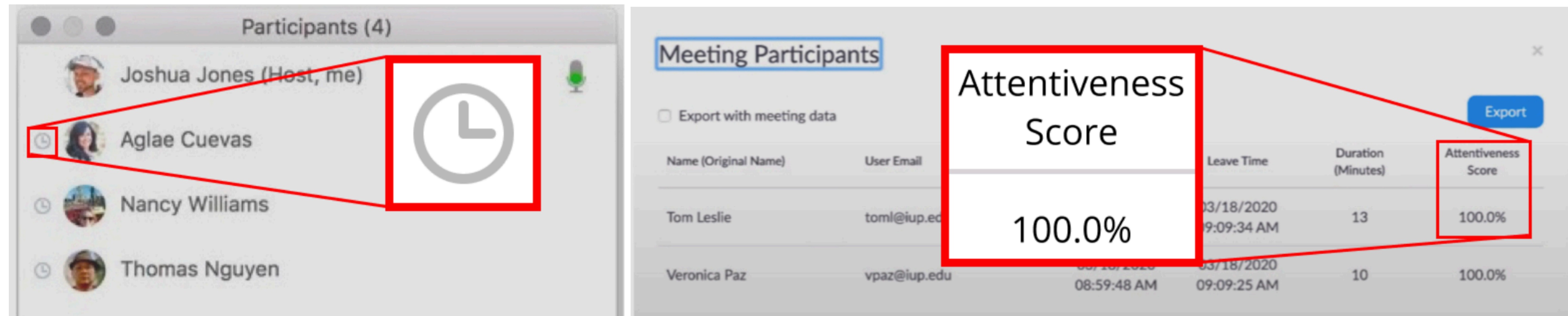
Close-ended Questions with Progressive Details

I want to design an attendee attention tracking feature for a video conferencing app.



Motivating Example

Zoom Attendee Attention Tracking



Key Privacy Design Decisions in Zoom Attention Case

- Opt-in or Opt-out?
- Only present aggregated data?
- When should we send the notification?
Just-in-time, Install time, or Periodically?
- Only enable the feature in the education setting?
- More salient notification? Always on?
- How to infer attention?
- Process data on the edge?
- Data retention? Where is the data stored?
- Who can access the data? Access control?
- Report attention prediction errors?
-



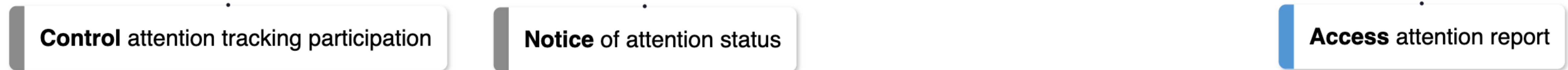
How to generate these questions?

Structure Representation of Privacy Design Decisions

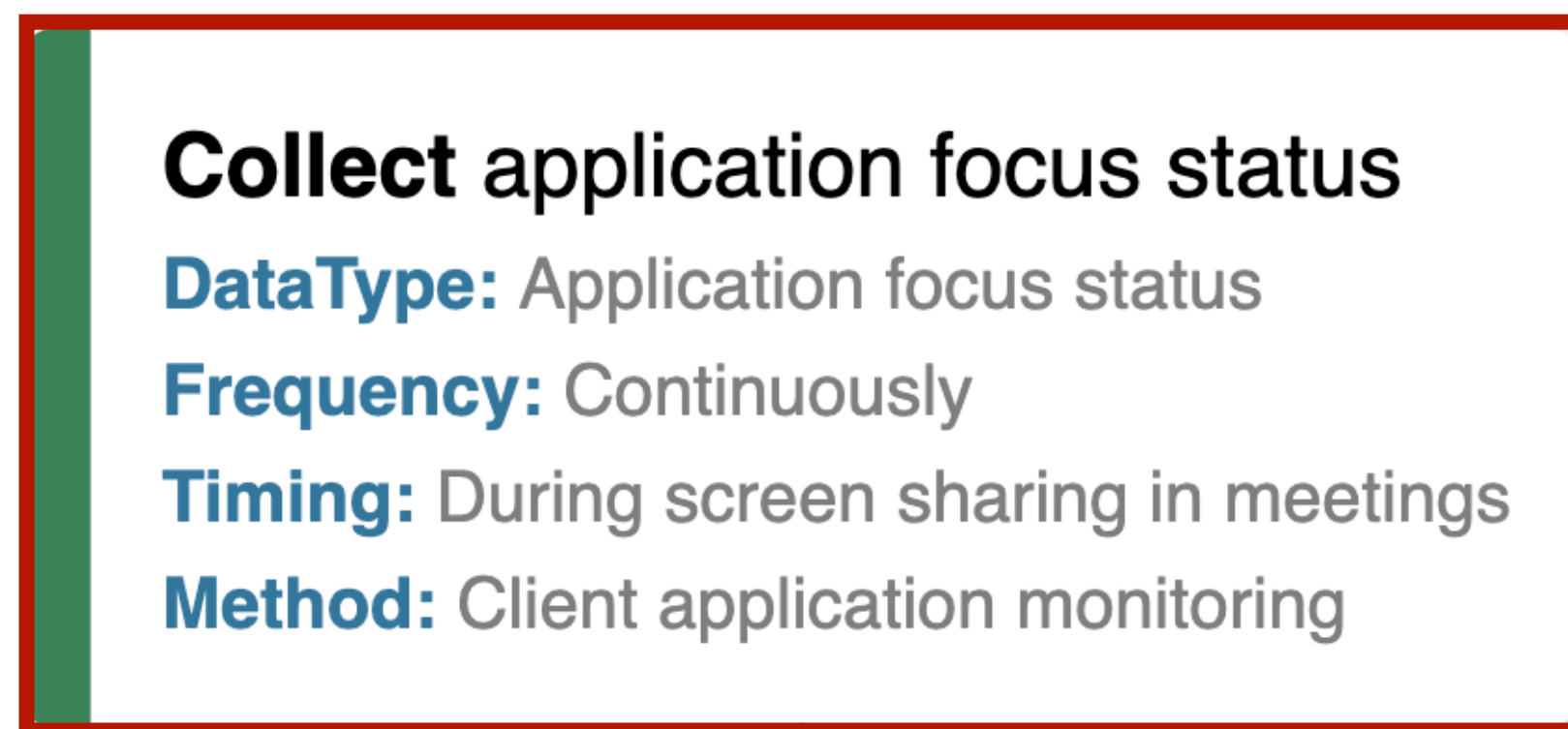
Layer 1: Data action



Layer 2: Stakeholder interaction



Layer 3: Individual design choices



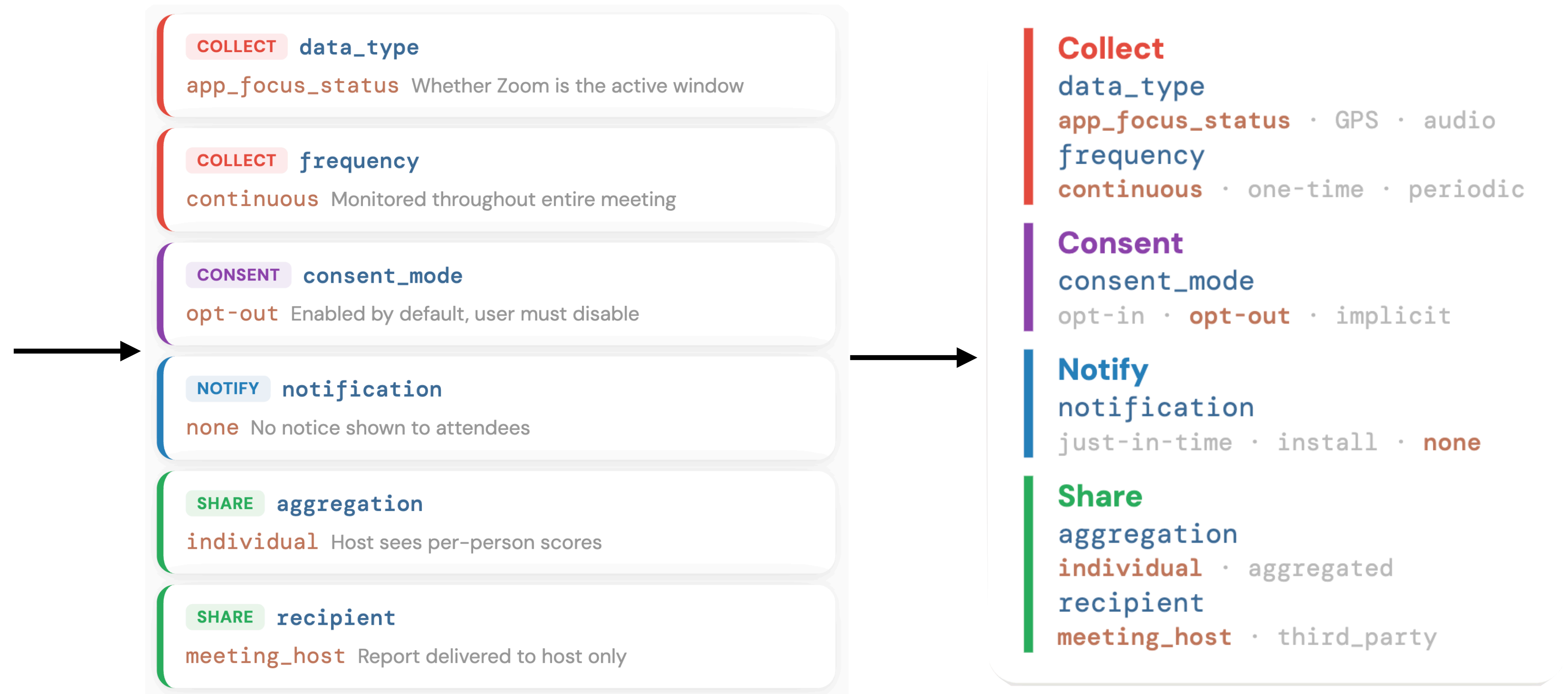
Method

Data-Driven Privacy Taxonomy

TECHCRUNCH

Zoom tracked whether attendees paid attention to screen shares

Zoom's attention-tracking feature monitored whether the app was the **active window** on participants' screens. The feature was **enabled by default** with **no notification** to attendees. Hosts received a **per-person attention score** in a post-meeting report.



10K Privacy news articles

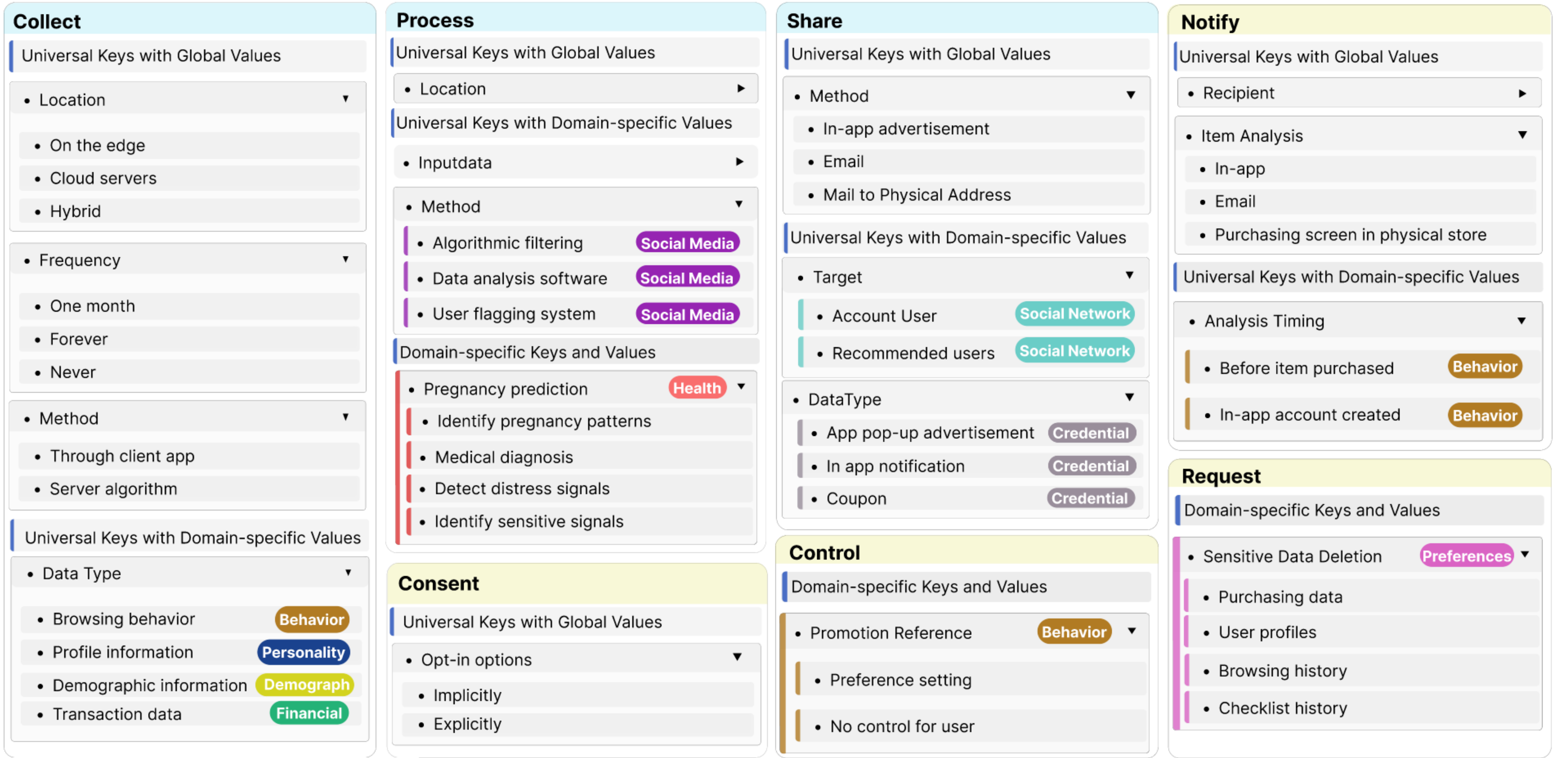
Extract design decisions

Iterative open coding

Results

Data-Driven Privacy Taxonomy

Domain Labels: **Health** **Social Media** **Demographic** **Financial** **Personality** **Social / personal Network** **Location** **Credential** **Behavior** **Preferences** **Data Actions** **Stakeholder Interactions**



Question Generation

Explorative question

Question 1

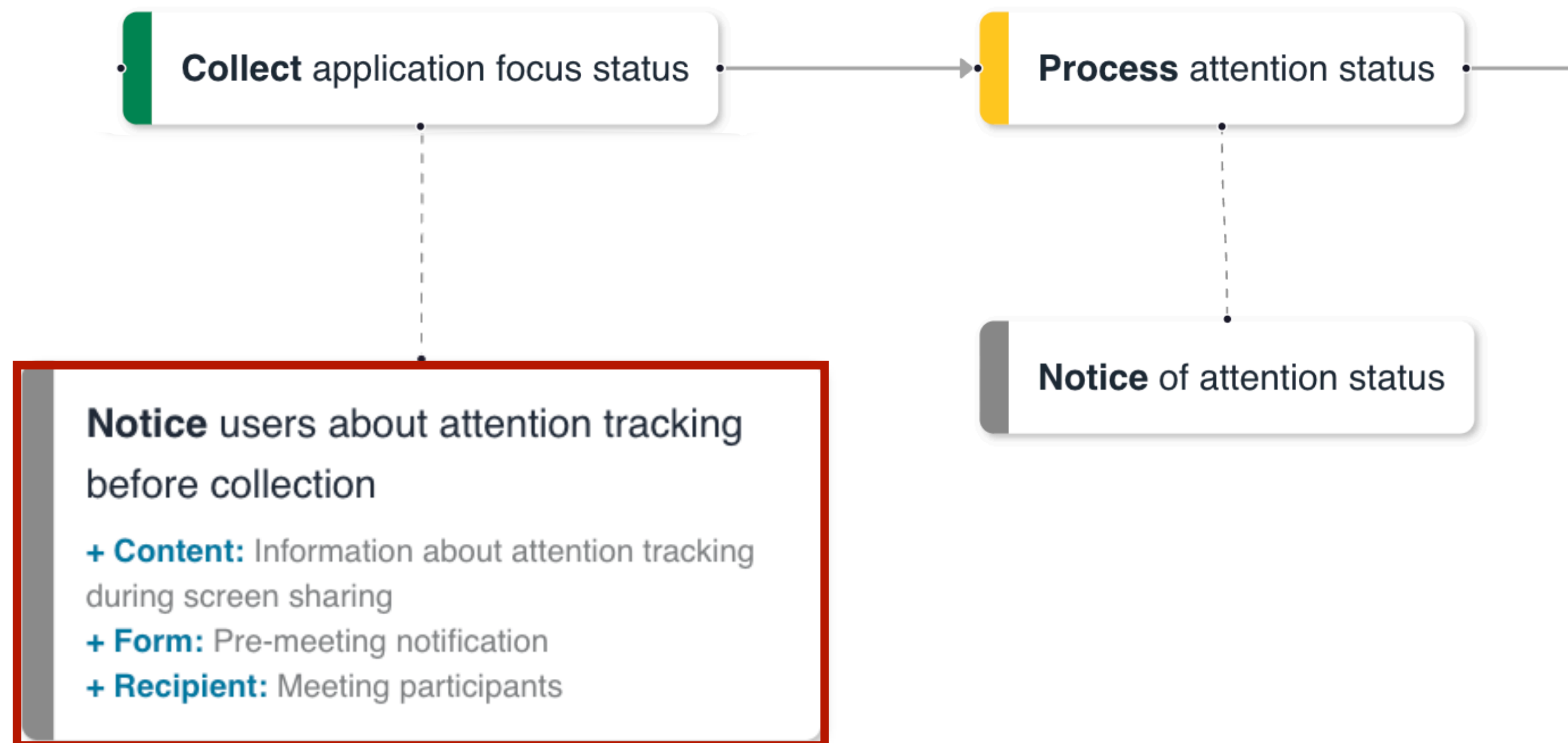
Would users be notified before their attention status is collected during screen sharing?

SELECT AN ANSWER [Write custom answer](#)

☒ Yes

☐ No

[< Previous](#) [Skip](#) [Next >](#)



Question Generation

Exploitative question

Question 6

The system stores attention status and percentage data after processing

How long would attention data be retained after processing?

SELECT AN ANSWER

Write custom answer

☐ During the active meeting

☐ 24 hours after meeting ends

☐ 7 days after meeting ends

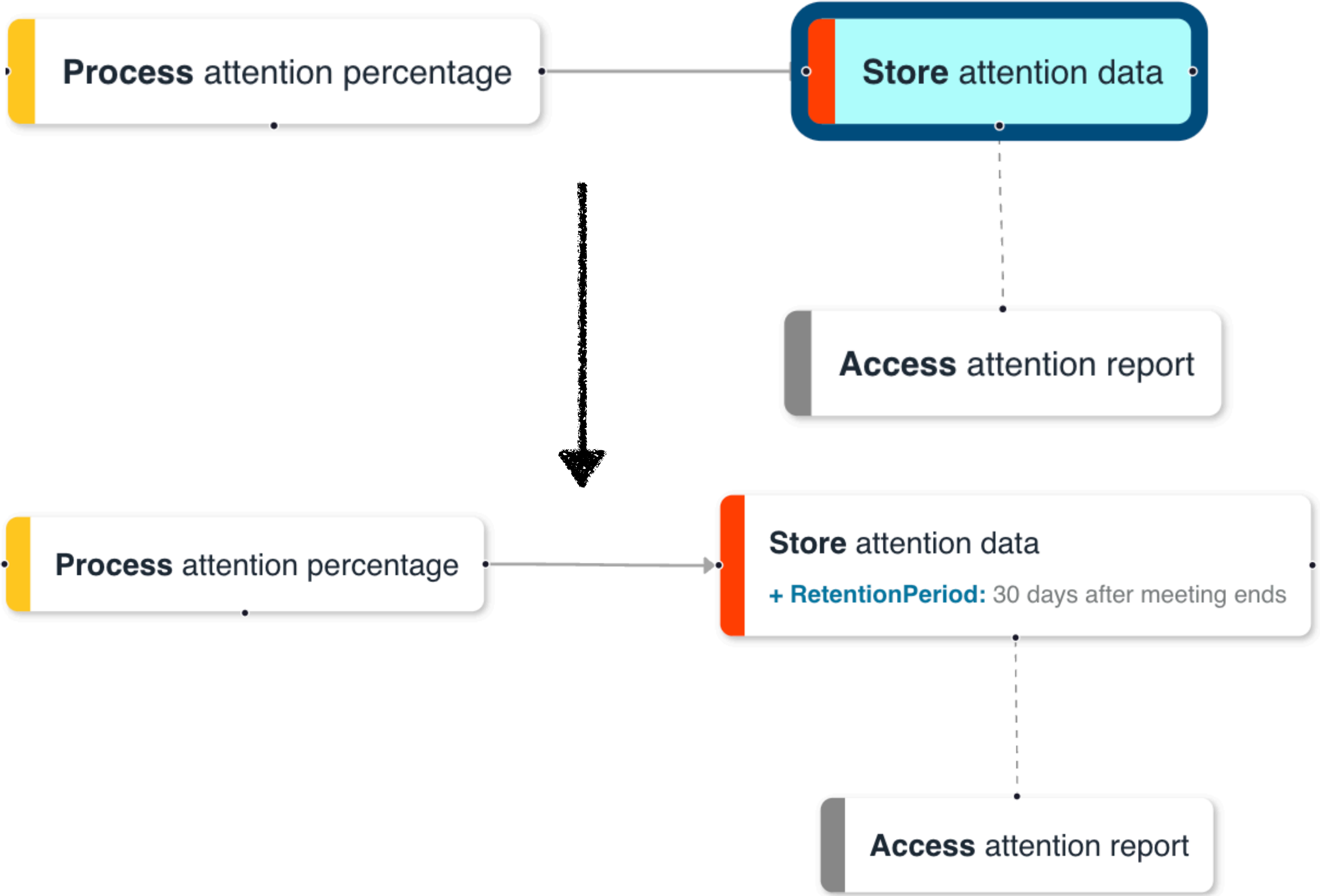
☒ 30 days after meeting ends

☐ Until manually deleted by admin

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What can we do with these questions?

Application Privacy Risk Assessment

① Input the design goal

Design an attendee attention tracking feature for a video conferencing application.

② Edit functional requirements

Data Collection

- Capture **application window focus state** from user's device
- Record **timestamps** when attention state changes

Data Processing & Storage

- Calculate the **percentage of attention time for each individual**
- Store processed attention metrics for **30 days**

User Interface and Presentation

- Display **a clock icon** next to inattentive participants
- Provide hosts with meeting summary of each participant

③ Answer LLM-generated questions

Question 1

Would users be notified before their attention status is collected during screen sharing?

Select an answer:

[Write Custom Answer](#)

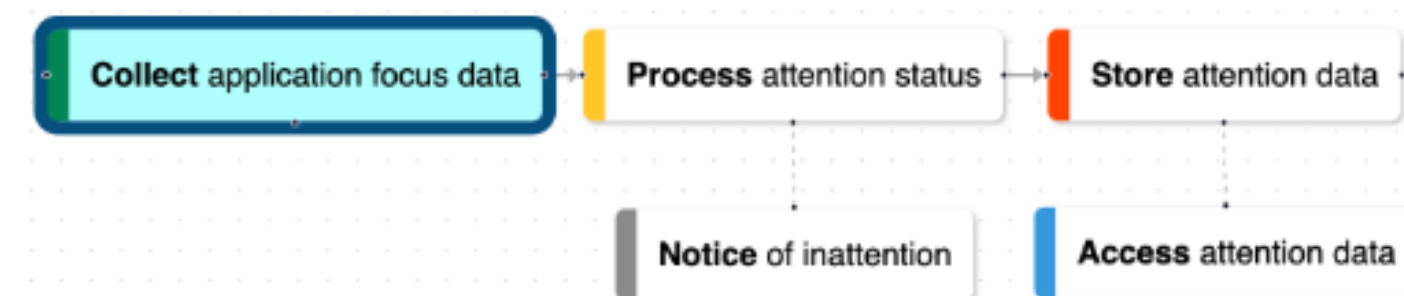
☒ Yes

☐ No

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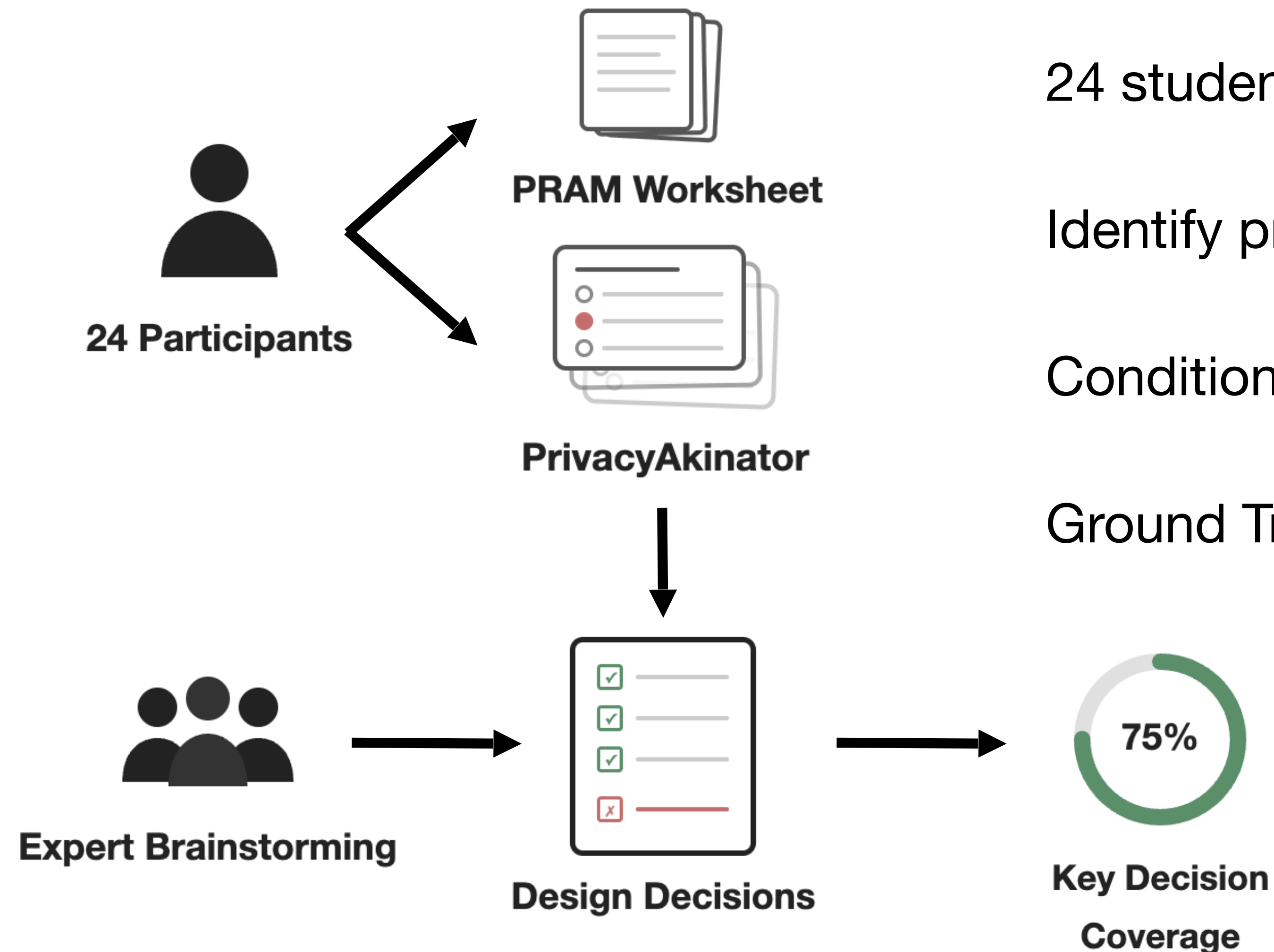


④ Assess system design and privacy risks

Data Actions	Summary Issues	Potential Problems for Individuals	Likelihood	Impact	Risk
Collect application focus status	Lack of multitasking exceptions may create false readings of inattention	Discrimination Loss of Trust	7	17	119
	Limited opt-out mechanism (account settings only) may not provide sufficient control at the point of use	Loss of Autonomy	6	20	120
Process attention status	Processing method doesn't distinguish between legitimate and non-legitimate reasons for switching applications	Dignity Loss Loss of Trust	2	17	34
Store attention data	Individual-level granularity increases privacy risk compared to aggregate data	Dignity Loss Loss of Trust	6	32	192
	30-day retention may exceed necessary period for feature functionality	Loss of Trust Economic Loss	5	19	95
Notice of attention status	Persistent indicator may create anxiety or behavior modification	Dignity Loss Loss of Autonomy	2	15	30
	Limited transparency about what specific behaviors are being monitored	Loss of Trust Loss of Autonomy	6	25	150
Access attention report	Participants have limited control over who can access their attention data	Loss of Trust Loss of Autonomy	7	36	252
Process data deletion	Questions about verification of complete deletion	Loss of Trust Economic Loss	5	10	50

Method

User Study



24 students & junior developers

Identify privacy-related design decisions

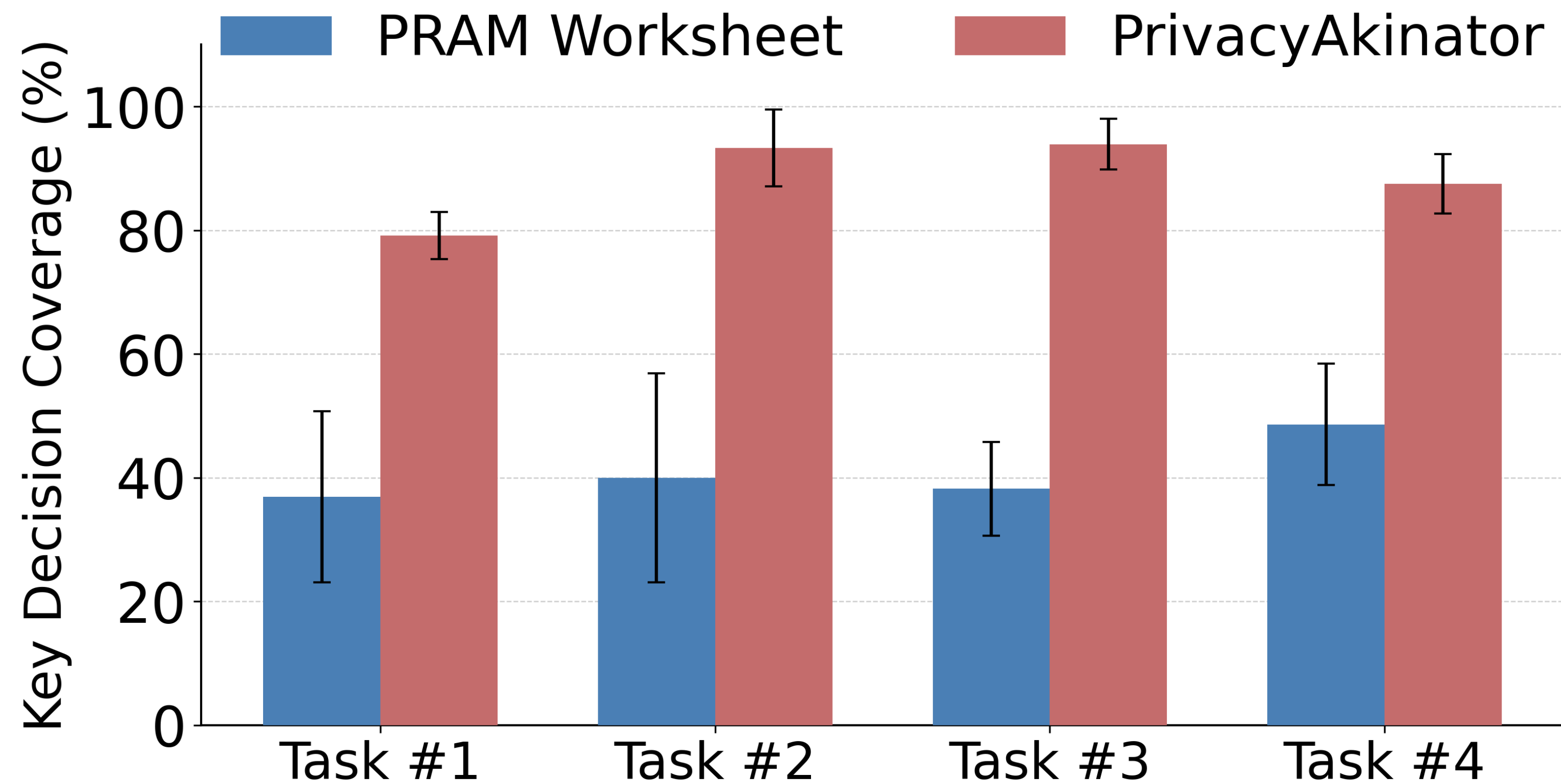
Conditions: PRAM worksheets vs. our tool

Ground Truth: Expert brainstorming

Results

User Study

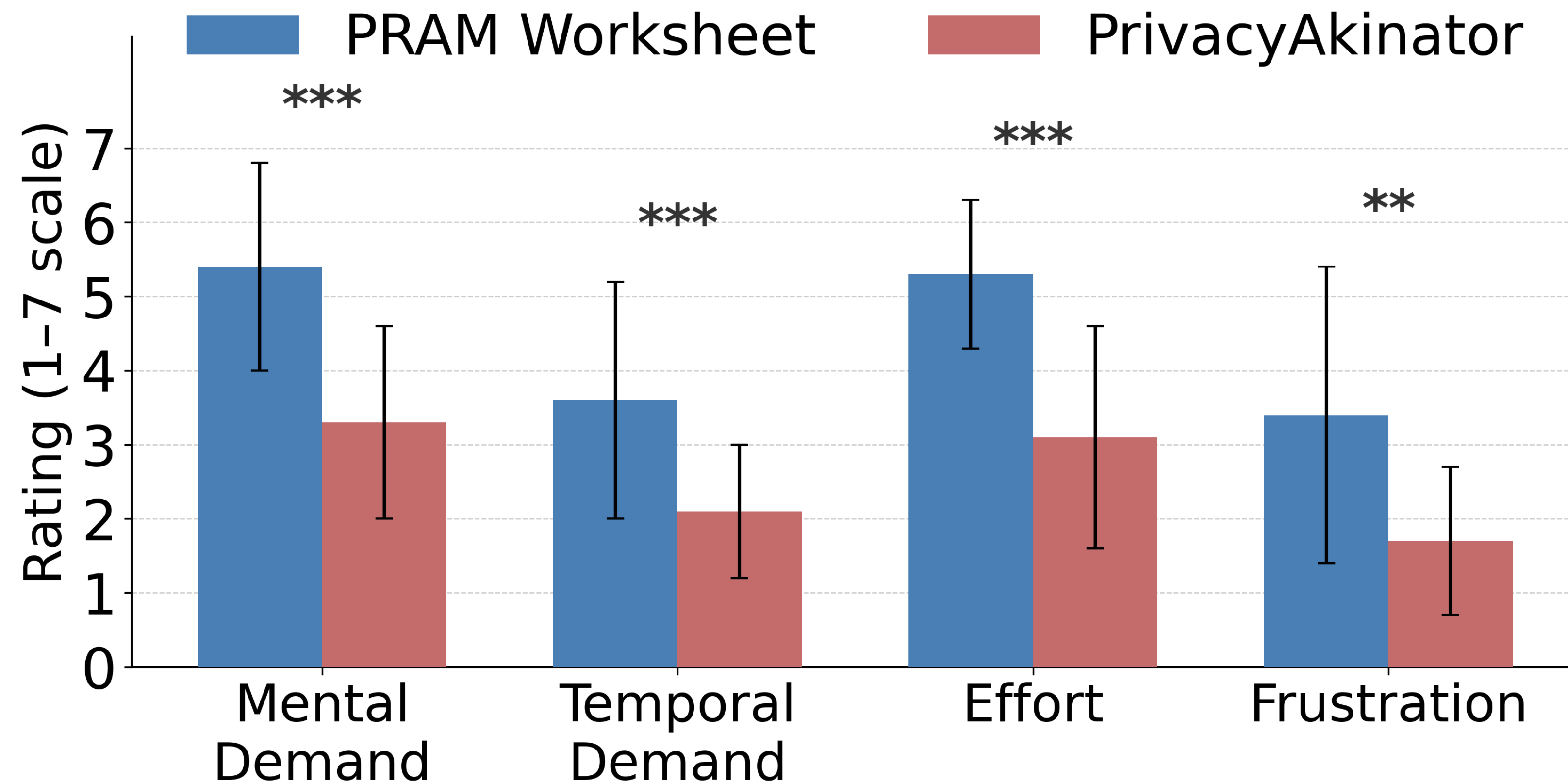
Coverage: **89% vs. 42%**



Results

User Study

Reduced cognitive load ($p < 0.01$)



PrivacyAkinator:

Decomposing Data Practice into Discrete Design Choices

I want to design an attendee attention tracking feature for a video conferencing app.



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